Soil Health Prediction from IoT Sensor Data Using Machine Learning in Sub-Saharan Africa: A Systematic Literature Review and Methodological Framework

Adrian Rugaba Asiimwe
Department of Computer Science, Uganda Christian University
Mukono, Uganda
Reg no: S24B23/054

***Abstract*—Soil degradation threatens agricultural productivity and food security, and many farmers in sub-Saharan Africa (SSA) have little access to laboratory soil testing, which is slow and requires specialised equipment. Internet of Things (IoT) field sensors combined with machine learning (ML) are a possible alternative: sensors measure soil properties such as moisture and pH in the field, and ML models estimate laboratory values from the sensor readings. This systematic literature review (SLR) examines how ML has been used to predict soil properties from field sensor data, how the models were validated, and how robust they are to noisy sensor inputs. It follows PRISMA 2020 and the guidelines of Kitchenham and Charters. Scopus and Google Scholar were searched, which returned 262 records; after duplicate removal, screening, full-text assessment and backward snowballing, 20 studies were included. No study that predicts soil properties from field sensor data with ML on Ugandan soils was found in the sources searched. The paper then sets out a methodological framework for an empirical study, with a sensor simulation, public soil datasets, and a comparison of random and spatially blocked validation.**

***Index Terms*—soil health, IoT sensors, machine learning, validation, precision agriculture, sub-Saharan Africa, Uganda**

I. Introduction

Soil degradation affects an estimated 1,036 to 1,470 million hectares worldwide and threatens long-term agricultural productivity and food security [1]. The range is wide because published estimates define degradation and assess it in different ways. Uganda's soils are reported to be highly degraded, low in nitrogen, phosphorus and potassium, and generally acidic [2]. Soil conditions vary between regions, so this finding applies to the areas covered by [2] and not necessarily to all of Uganda.

Farmers can respond only if they know what their soil lacks, yet limited access to soil testing services has left many farmers in districts such as Nakapiripirit and Amudat struggling with declining fertility [3]. Without testing, farmers judge fertility by vegetation cover, weeds, soil colour and previous yield. Elaete et al. compared these four indicators with Palin test-kit results for 461 beneficiaries of an input-subsidy project in Uganda. Depending on the indicator, between 61 percent (previous yield) and 71 percent (vegetation cover) of farmers misclassified their soil fertility [2].

Laboratory analysis remains the scientific standard for soil health assessment, but it is slow and out of reach for many farmers. Typical procedures include atomic absorption spectrometry, the Kjeldahl method for total nitrogen, and spectrophotometric analysis for boron and phosphorus [1]. All require specialised equipment, trained personnel and controlled conditions that are seldom available outside major research institutions. Visual indicators are unreliable and laboratory tests are hard to reach, so farmers need an automated, field-based method that returns nutrient-specific estimates at many points on a farm.

Field sensors combined with ML offer such a method. Sensors measure soil moisture, pH and nutrient levels in real time [4]. Moisture determines water availability, and pH controls how much of each nutrient plants can absorb [5]. ML models trained on paired sensor and laboratory data can estimate nutrient levels directly from sensor readings; the properties predicted include nitrogen, phosphorus and potassium [4], [6]. Reviews identify soil sensors as the most frequently employed technology for soil health assessment [7], and portable spectral analysis with edge artificial intelligence has yielded near-laboratory accuracy for in-situ measurement [10], [11].

Farming context. About 7 million Ugandan households, 80 percent of the total, cultivated land or reared livestock in 2019, with an average holding of 1.3 ha [15]. The 2024 census reports that 17 percent of households engage in commercial agriculture, from 27 percent in Bunyoro and Buganda to 7 percent in Karamoja [16]. The sources found do not describe the crops or soil management of commercial farms; the 2008/09 Census of Agriculture used a separate form for large-scale and institutional farms [17]. The review therefore records farm type (smallholder or commercial) for each included study.

A. Problem Statement and Research Gap

Ugandan research has compared farmers' indicators with test-kit results and called for affordable, portable testing, but it has not built or evaluated a predictive model [2]. Sensor-based ML studies exist elsewhere [4], [6], [11], but the literature reviewed here leaves three questions insufficiently examined:

(i) whether such models work on Ugandan soils, (ii) how the models were validated, since random cross-validation on spatially clustered samples can look more accurate than it will be on new fields, and (iii) how robust they are to the noise expected in field sensor data. Section VII states each gap from the reviewed studies.

B. Aims and Research Questions

The review is guided by three research questions.

- **RQ1:** Which IoT or field sensors and soil parameters are used in soil health monitoring, and how well do their readings agree with laboratory values? *Measured by:* sensor types and parameters per study, agreement with laboratory values (R^2, RMSE or bias), and the deployment challenges reported (connectivity, power, calibration and maintenance, skills, data availability, usability).
- **RQ2:** Which machine learning models are most commonly used to predict soil properties from sensor data, and how do they perform in terms of accuracy and data requirements? *Measured by:* the number of included studies using each model family (a study counts once per family), reported R^2, RMSE or accuracy compared only between studies reporting the same metric for a similar task, and dataset sizes.
- **RQ3:** How were the ML models validated, and how robust are they to noisy sensor inputs? *Measured by:* the validation type (independent or spatially blocked data, random split, or cross-validation only) and whether input noise or sensor error was tested.

The objectives are to review field sensing of soil properties, identify the ML models used to predict them, examine how those models were validated and tested for noise, and identify the gaps that support a methodological framework for an empirical study on Ugandan farms.

C. Scope

GIS mapping was considered during planning and removed from the scope. The studies reviewed did not provide a basis for justifying maps as a component of the prediction problem, and mapping adds interpolation choices that are separate from the ML question. Spatial information is used here only to design validation, by holding out whole areas. Economic analysis is also outside the scope, so the review does not classify approaches by cost.

II. Method

A. Research Approach and Protocol

This study is an SLR. A systematic approach makes the search, selection and analysis of studies explicit and repeatable, which reduces selection bias compared with a narrative review. The review was planned and is reported following the PRISMA 2020 statement [12], supported by the SLR guidelines of Kitchenham and Charters [13] and the PRISMA-S extension for reporting searches [14]. The protocol sets out the research questions, databases, search strings, criteria, quality assessment items and data extraction form. The review proceeds in four stages: identification, screening, eligibility assessment and final inclusion (Fig. 1).

TABLE I
Databases Searched and Records Identified.

Database	Fields searched	Records
Scopus	Title, abstract, keywords	112
Google Scholar	Title and full text; up to the first 100 results per query, sorted by relevance	150
Total		262

Operational definitions. "Farmers" means people who grow crops or manage agricultural land, whatever the size of their holding.

Sensor. A sensor is a field-deployable or proximal device that measures a soil property without laboratory processing (for example electrochemical, capacitive, optical, spectral or laser-induced breakdown spectroscopy devices). Laboratory instruments and airborne or satellite imagery are not counted.

Evidence origin. Each included study is labelled *SSA-demonstrated* (data collected in SSA) or *transferable* (data from another region, with the authors stating that the method applies to SSA). Findings are reported separately for the two labels.

Validation types. Independent validation tests on data from separate fields, sites or seasons. *Spatially blocked* validation holds out whole areas. *Random split or cross-validation* divides samples at random, so near neighbours can appear in both training and test sets.

B. Databases Searched

Two sources were searched (Table I). Scopus was chosen as the main source for its multidisciplinary index with strong agricultural, environmental and engineering coverage. Google Scholar was added as a supplementary source for studies missed by Scopus, including regional work. Web of Science and ScienceDirect were not searched; this is acknowledged as a limitation (Section VIII). Google Scholar results change over time and cannot be exported directly, so they were retrieved with Publish or Perish. Each query was sorted by relevance and limited to the first 100 results, patents were excluded, and the pooled results of all queries gave 150 records after citation-only entries were removed.

C. Search Keywords and Search Strings

Keywords were derived from the research questions and fall into three concept groups: soil, technology and context. Synonyms within a group are joined with OR, and the three groups with AND. The master Boolean string is:

```
("soil health" OR "soil fertility" OR
"soil quality" OR "soil nutrient*" OR
"soil moisture" OR "soil pH" OR
"soil organic carbon") AND
("machine learning" OR "deep learning"
OR "predictive model*" OR "IoT" OR
"Internet of Things" OR
"wireless sensor*" OR "soil sensor*"
OR "spectroscopy" OR "edge AI")
```

TABLE II
INCLUSION CRITERIA.

ID	Inclusion criterion
IC1	Peer-reviewed journal article or conference paper
IC2	Published from January 2015 onward
IC3	Written in English
IC4	Addresses soil health, fertility or quality (e.g. nutrients, pH, moisture, organic matter)
IC5	Uses an IoT or field sensor together with ML to predict or estimate a soil property
IC6	Set in SSA; or set in another tropical or low-resource region *and* the authors explicitly discuss transfer to SSA (e.g. comparable soils, climate or farming systems)
IC7	Reports enough technical detail (sensors, models, data) to allow data extraction

TABLE III
EXCLUSION CRITERIA.

ID	Exclusion criterion
EC1	Not peer-reviewed (e.g. theses, editorials, white papers, posters)
EC2	Published before 2015 or not in English
EC3	Focuses on topics other than soil, such as crop disease, weather or livestock only
EC4	Does not combine sensor data with ML (fails IC5)
EC5	Opinion or position paper with no technical or empirical content
EC6	Duplicate or earlier version of a study already included
EC7	Full text not accessible
EC8	Outside SSA and not applicable to it (fails IC6)
EC9	Insufficient technical detail for data extraction (fails IC7)

```
AND ("farmer*" OR "farming" OR
"precision agriculture" OR
"sub-Saharan Africa" OR "East Africa"
OR "Africa" OR "Uganda" OR "Kenya" OR
"Tanzania" OR "Ethiopia" OR "Rwanda"
OR "Nigeria" OR "Ghana")
```

The technology group deliberately joins the sensor and ML terms with OR so that studies using only one of them are retrieved; the requirement of both (IC5) is applied during screening. Remote sensing is not used as a search term, because the review concerns sensors that farmers or extension workers can operate in the field. The context group contains general farming terms, so studies from other regions are retrieved and then judged for applicability to SSA under IC6.

In Scopus the string is wrapped in TITLE-ABS-KEY(...). Google Scholar does not support long Boolean strings, so the string was split into shorter queries, each combining one term from each concept group, and the results were pooled.

D. Inclusion and Exclusion Criteria

A study must meet all inclusion criteria (Table II) and none of the exclusion criteria (Table III). The start date of 2015 focuses the review on the period in which low-power IoT hardware and applied ML became widely used in agriculture. IC6 admits studies from other regions only when the authors discuss transfer to SSA, not merely because they mention smallholders or the tropics.

TABLE IV
DATA EXTRACTION FIELDS.

Category	Data extracted	RQ
Study details	Authors, year, title, venue, country or region, crop or farming system, scale, farm type (smallholder or commercial), evidence origin (SSA-demonstrated or transferable)	All
Sensing	Sensor types, soil parameters, power source, connectivity, calibration, deployment setting, agreement with laboratory values	RQ1
Deployment	Reported challenges (connectivity, power, calibration and maintenance, skills, data availability, usability) and proposed solutions	RQ1
Machine learning	Algorithm(s) and model family, input features, dataset size, metrics (e.g. accuracy, R^2, RMSE)	RQ2
Validation	Validation type (independent set, spatial blocks, random split or cross-validation), and whether robustness to input noise was tested	RQ3
Findings and limitations	Main results, stated limitations and gaps, quality score	All

E. Screening, Selection and Duplicate Removal

Selection followed six stages: (1) identification, in which all records from the database searches were exported and combined; (2) duplicate removal; (3) title and abstract screening, where clearly failing records were excluded and uncertain ones moved on; (4) full-text assessment, where the reason for each exclusion was recorded with codes EC1 to EC9; (5) backward snowballing of the reference lists of included studies, with the same screening steps; and (6) final inclusion of studies from databases and snowballing for quality assessment, extraction and synthesis.

Screening decisions and exclusion reasons were recorded in a screening log. Records were exported in RIS or BibTeX format into a reference manager. Duplicates were detected first by matching DOIs, then by matching title, first author and publication year. Possible matches were checked manually before deletion, and the version with the most complete metadata was kept.

Single-reviewer screening. The author screened all records, so no second reviewer was used and inter-rater agreement was not measured. This is the main source of selection bias in this review, because one person's reading of the criteria decided every inclusion and exclusion. Every decision and its reason was logged so that a second reviewer can check it.

F. Data Extraction

Data were recorded in a standardised spreadsheet form, which was first tested on a small sample of included studies and then refined before full extraction. Fields were chosen so that every research question can be answered (Table IV). Information a study does not report is recorded as "not reported" instead of being estimated.

TABLE V
QUALITY ASSESSMENT CHECKLIST (MAXIMUM 6 POINTS).

Item	Question	Scoring
Q1	Are the aim and farming context clearly stated?	1 yes; 0 no
Q2	Are the data source and sample size described?	1 both; 0.5 one; 0 neither
Q3	Is the method detailed enough to repeat?	1 yes; 0.5 partly; 0 no
Q4	Was the result validated on independent data?	1 independent set (separate field, site or season) or spatially blocked validation; 0.5 random split or k-fold cross-validation only; 0 none. For sensor-only studies: validation against laboratory values
Q5	Are performance metrics reported?	1 yes; 0.5 partly; 0 no
Q6	Are limitations discussed?	1 yes; 0.5 briefly; 0 no

G. Quality Assessment

Each study receives a quality score out of 6 using the checklist in Table V. The six items cover the main stages of a study (aim, data, method, validation, results and reflection), in line with the general quality concerns discussed in [13]. They carry equal weight because no agreed weights exist for studies that mix sensing and ML, and unequal weights would add subjective judgement; a sensitivity check (below) tests whether this choice affects the findings.

Validation (Q4) distinguishes testing on independent data from cross-validation. Random k-fold cross-validation reuses data from the same samples and can overstate accuracy on new fields, so it earns half a point; an independent dataset or spatially blocked validation earns a full point.

The score is not used to exclude studies. Studies scoring below 3 are retained but (i) are marked in every table, (ii) are left out of the numeric ranges for accuracy, and (iii) are cited in the synthesis only as supporting evidence beside higher-scoring studies. Each finding is also reported with and without these studies, so the effect of the weighting can be seen. Because studies are expected to differ in sensors, models, datasets and metrics, no meta-analysis is planned, and performance figures are compared only between studies reporting the same metric for a similar task.

III. SELECTION OF STUDIES

The searches identified 262 records (Scopus 112, Google Scholar 150). After 31 duplicates were removed, 231 records were screened by title and abstract, and 174 were excluded. Of the 57 reports sought for retrieval, 6 could not be retrieved (EC7) and 51 were assessed in full text. Full-text assessment excluded 34 reports (EC1–EC2: 8; EC3: 9; EC4: 10; EC8: 4; EC9: 3), leaving 17 studies. Backward snowballing of their reference lists added 3 studies, giving 20 studies for the synthesis (Fig. 1).

Search yield. The string requires a term from each of three concept groups, so the yield was limited to studies that mention soil, a technology and a farming context together. Section VIII discusses the effect of this on coverage.

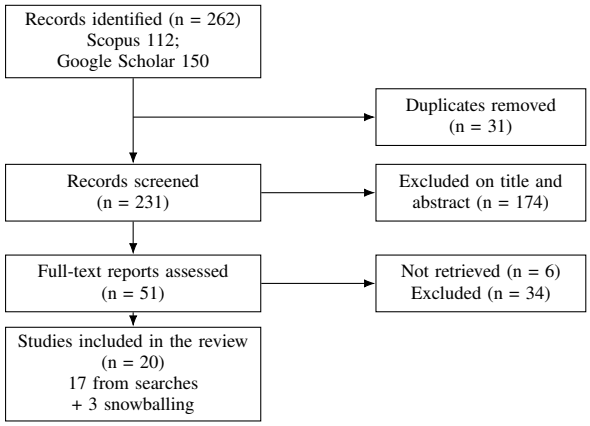


Fig. 1. PRISMA 2020 flow diagram of the study selection process. Reasons for full-text exclusion are given in Section III.

IV. INCLUDED AND BACKGROUND STUDIES

Table VI compares the background studies cited in this paper on three dimensions: field sensing, ML-based prediction and evaluation in SSA. Background studies give context for the research gap but are not part of the synthesis in Section V; any statement that rests only on background studies is marked "Background". By the criteria in Table II, studies [1], [2], [10] and [9] use a sensor or ML but not both, and are therefore background only. None of the studies in Table VI covers all three dimensions, whereas the proposed direction does.

V. RESULTS BY RESEARCH QUESTION

This section synthesises the 20 included studies by research question. Statements marked "Background" come from the background studies in Table VI and are not findings of the review.

A. RQ1: Sensors and soil parameters

The included studies are grouped by sensor type and soil parameter measured. Agreement with laboratory values is compared only between studies reporting the same metric for the same property, and results are reported separately for SSA-demonstrated and transferable studies and for smallholder and commercial farms. Deployment challenges (connectivity, power, calibration and maintenance, skills, data availability, usability) are counted across studies. *Background:* field sensors measuring soil moisture, pH and nutrient levels in real time are the basis of IoT soil monitoring [4], and reviews report soil sensors as the most frequently used technology for soil health assessment [7].

B. RQ2: Machine learning models

Models are grouped into families (for example random forest, support vector machine, neural network and regression), and a study counts once per family. Accuracy (R^2, RMSE or classification accuracy) is compared only between studies

TABLE VI
BACKGROUND STUDIES AND THE PROPOSED DIRECTION. SENSOR FOLLOWS THE DEFINITION IN SECTION II-A.

Study	Main contribution / limitation	Sensor	ML	SSA
Elaete et al. [2]	Farmer indicators vs Palin test-kit results in Uganda; no predictive model	No	No	Yes
Abdullah et al. [9]	IoT nutrient monitoring system, tested in Indonesia	Yes	No	No
Islam et al. [4]	ML-enabled IoT system for soil nutrient monitoring	Yes	Yes	No
Shwetha and Bhat [6]	Sensor-based deep learning for soil nutrient prediction	Yes	Yes	No
Chlouveraki et al. [1]	Soil properties from hyperspectral imaging and ML	No	Yes	No
Erler et al. [10]	Mobile laser-induced breakdown spectroscopy; field-deployable, so a sensor under Section II-A	Yes	No	No
Kasemi [11]	Soil sensor with edge AI in agricultural robotics	Yes	Yes	No
Mamabolo et al. [7]	Review of precision agriculture technologies	Review	Review	No
Getahun et al. [8]	Systematic review of precision agriculture technologies	Review	Review	No
Proposed direction	Sensor-based ML prediction with spatial validation and noise tests on Ugandan soils	Yes	Yes	Yes

reporting the same metric for a similar task, and studies scoring below 3 are left out of the ranges. Dataset sizes are recorded because they limit which models can be trained. *Background:* ML and deep learning models trained on paired sensor and laboratory data estimate nutrient levels from sensor readings [4], [6].

C. RQ3: Validation and robustness

Each ML study is classed as independent, spatially blocked, or random-split or cross-validation only, and checked for whether it tested robustness to input noise or sensor error. The counts show how much of the reported accuracy rests on validation that can overstate performance on new fields.

VI. DISCUSSION

Laboratory analysis is the standard for soil health assessment but is out of reach for many farmers [1], [3]. Sensors and ML each address part of the problem [4], [6], and edge AI shows that near-laboratory accuracy is possible with portable tools [10], [11]. The reviewed literature therefore supports predicting soil properties from field sensor data with ML and evaluating the models against laboratory results on African soils.

Strength of the evidence. The final evidence base is 20 studies, from two sources and snowballing, screened by one reviewer. Findings from it show where the literature points and do not prove that a gap exists. The effect is larger where a finding rests on few studies or on studies scoring below 3, and each finding in Section V is reported with and without those studies for that reason.

Machine learning reliability. For ML-based soil prediction, the validation design affects how far an accuracy figure can be trusted. Soil samples from nearby locations are similar, so random splits place near-duplicates in both training and test sets and can overstate accuracy; spatially blocked validation, which holds out whole areas, gives a harder and more realistic test of performance on a new farm. Field sensors add a second problem, because their readings are noisier than laboratory values, so a model that performs well on clean data may degrade quickly on field data. These points are why the review records the validation type (Table V, Q4) and any noise test for every ML study.

Alternative explanations for the gap. The absence of Ugandan work that predicts soil properties from field sensor data with ML may mean that none exists. It may also reflect (i) work published as theses, reports or in journals that the two sources do not index (excluded here under EC1); (ii) different terminology, such as "rapid soil test" or "soil spectroscopy", which the string only partly covers; (iii) language limits, since only English-language studies were included; and (iv) timing, since low-power hardware and applied ML are recent in this setting. The gap is therefore best stated as "not found in the sources searched".

Transferability. Results from outside SSA are labelled transferable and should not be read as SSA evidence. Sensor readings and ML calibrations can depend on soil type, moisture and temperature, so a model trained elsewhere may not hold on Ugandan soils until tested there.

VII. RESEARCH GAPS

The gaps are stated from the reviewed studies. Among the seven primary studies in Table VI (excluding the two reviews): (1) one was set in SSA [2], and it built no predictive model; (2) three combined sensors with ML [4], [6], [11], none of them on Ugandan soils; and (3) whether such models were validated on spatially independent data and tested against noisy inputs is a question the included studies are examined for through the validation and noise-test fields in Table IV.

VIII. LIMITATIONS

Only two sources were searched. Web of Science, ScienceDirect and African and agricultural indexes such as AJOL and AGRIS were not used, so relevant publications elsewhere may be missed; extending the search to them is the first step for the next version. Google Scholar results were limited to the first 100 per query. Only English-language studies from 2015 onward were included, which excludes French- and Portuguese-language work from SSA. Remote-sensing-based soil mapping was outside the search, so studies using satellite or drone imagery may be missed. Screening was done by a single reviewer, which raises the risk of selection bias, and no inter-rater agreement was measured. GIS mapping and economic analysis are outside the scope, so the review says nothing about them. The sources found also do not describe

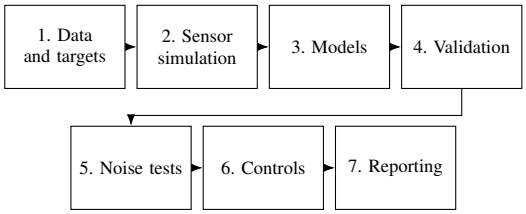


Fig. 2. Methodological pipeline for the proposed study.

TABLE VII
CANDIDATE PUBLIC DATASETS.

Dataset	Content	Source
Open Soil Spectral Library	91,631 mid-infrared and 65,063 visible-near-infrared scans with laboratory data from more than 60,000 locations; includes AfSIS and ICRAF-ISRIC libraries and a portable-instrument NIR subset scanned on 9 devices	[18]
Africa Soil Profiles Database v1.2	18,532 legacy profiles for SSA; analytical data for 15,564, of which 14,197 are georeferenced; profile reliability rated 1–4	[19]
WoSIS snapshot (2019)	196,498 georeferenced profiles from 173 countries, including organic carbon and pH	[20]

land use by commercial farmers in detail, so conclusions about commercial farms rest on the farm-type labels in the included studies. Finally, the gaps reported here show limits of the evidence found and are not proof that no study exists.

IX. METHODOLOGICAL APPROACH FOR THE PROPOSED EMPIRICAL STUDY

The proposed study answers the validation and robustness questions directly. It has two parts, which share one modelling pipeline: an analysis of public soil datasets (Part B) and a simulation of field sensors (Part A). Fig. 2 shows the stages, and the steps below define each one.

Step 1: data and targets. The target properties are soil pH and soil organic carbon (SOC), with moisture as a covariate. Part B uses public datasets that hold coordinates, laboratory values and either spectra or covariates (Table VII). A dataset is selected if it has coordinates and laboratory SOC or pH, a documented licence, and samples from Africa. The Ugandan subset is counted and reported separately. Records with missing coordinates or targets are removed, units are harmonised, duplicates are dropped, and the dataset version is recorded.

Step 2: sensor simulation (Part A). A synthetic field with spatially correlated pH, SOC and moisture is surveyed at a fixed number of points. Each "sensor" reading is the true value passed through a measurement model with gain, offset, random noise, calibration drift and cross-sensitivity to moisture. Parameters come from sensor datasheets or bench tests, and any value without a source is labelled an assumption. The model is run at several noise multipliers.

Step 3: models. Three families are compared: a regularised linear or PLS regression baseline, random forest, and gradient boosting. Hyperparameters are tuned by nested cross-validation inside the training folds only, and random seeds are fixed.

Step 4: validation. Every model is evaluated twice on the same data: with random 5-fold cross-validation and with spatially blocked cross-validation, where samples are grouped by clustering their coordinates and whole groups are held out. Performance is reported as R^2, RMSE and the ratio of performance to interquartile distance (RPIQ):

$$\mathrm{RMSE} = \sqrt{\tfrac{1}{n}\textstyle\sum_i (y_i - \hat{y}_i)^2}, \quad \mathrm{RPIQ} = \frac{\mathrm{IQR}(y)}{\mathrm{RMSE}}. \qquad (1)$$

The validation gap is the R^2 under random folds minus the R^2 under spatial blocks. Repeating the procedure over multiple seeds gives a spread for each figure.

Step 5: noise tests. Gaussian noise is added to the inputs at levels from 0 to 100 percent of each feature's standard deviation. Models are tested with clean training and noisy test inputs, and with noisy training and noisy test inputs. For Part A the noise levels follow the simulated sensor model; for Part B they are compared with instrument-to-instrument variation in the portable-instrument NIR subset of the Open Soil Spectral Library.

Step 6: controls. A model trained on shuffled targets must score near zero R^2, and a mean-value predictor sets the lower baseline. The block size used for spatial validation is varied to check that conclusions do not depend on it.

Step 7: reporting. Code, seeds and dataset versions are released with the results. Results from Part A are reported as simulations under stated assumptions and are not claims about real sensors. Results from Part B are reported for the subset and region actually used.

X. HYPOTHESES FOR THE PROPOSED EMPIRICAL STUDY

This section is separate from the review findings. The hypotheses follow from the research gap and the machine learning concerns above, and will be tested with the methodology in Section IX. They are not results of this review.

- **H1 (validation design):** On the same public dataset, SOC models score lower under spatially blocked validation than under random k-fold cross-validation.
- **H2 (noise robustness):** Prediction accuracy falls as input noise rises, and the rate of decline differs between model families.
- **H3 (sensor feasibility):** For soil pH, models using simulated field sensor inputs keep useful skill under spatial validation up to a stated noise level, whereas SOC is not predicted well from the sensor channels without spectral inputs.

XI. CONCLUSION

This review examined how ML has been used to predict soil properties from field sensor data among farmers in SSA. Of the 262 records identified, 20 studies were included. In the

sources searched, no study that predicts soil properties from field sensors with ML on Ugandan soils was found. The review treats validation design and noise robustness as the central questions for such models. The evidence base is small, so the next steps are to widen the search, add a second screener, and carry out the empirical study described in Section IX.

ACKNOWLEDGMENT

The author thanks Uganda Christian University and the Department of Computer Science for academic support during this study.

REFERENCES

[1] E. Chlouveraki *et al.*, "Estimation of soil properties using hyperspectral imaging and machine learning," *Smart Agric. Technol.*, vol. 10, art. no. 100790, 2025.

[2] S. Elaete, J. Ilukor, R. E. Isoto, A. Moruleng, and H. N. Opolot, "Comparing indigenous and scientific (Palin test) soil fertility assessment approaches for better soil health monitoring, management, and farm decision-making in Uganda," *Discover Soil*, vol. 2, art. no. 29, 2025, doi: 10.1007/s44378-025-00058-z.

[3] Food and Agriculture Organization of the United Nations, Global Soil Partnership, "Uganda rolls out the Global Soil Doctors Programme to bring soil health closer to farmers in Karamoja." [Online]. Available: https://www.fao.org/global-soil-partnership/resources/highlights/detail/en/c/1757615/

[4] M. R. Islam, K. Oliullah, M. M. Kabir, M. Alom, and M. F. Mridha, "Machine learning enabled IoT system for soil nutrients monitoring and crop recommendation," *J. Agric. Food Res.*, vol. 14, art. no. 100880, 2023, doi: 10.1016/j.jafr.2023.100880.

[5] D. Neina, "The role of soil pH in plant nutrition and soil remediation," *Appl. Environ. Soil Sci.*, vol. 2019, art. no. 5794869, 2019, doi: 10.1155/2019/5794869.

[6] G. N. Shwetha and G. J. Bhat, "Sensor based framework for soil nutrients prediction using deep learning techniques," *Agric. Eng. Int.: CIGR J.*, vol. 27, no. 4, pp. 291–301, 2025.

[7] E. Mamabolo *et al.*, "Application of precision agriculture technologies for crop protection and soil health," *Smart Agric. Technol.*, vol. 12, art. no. 101270, 2025.

[8] S. Getahun, H. Kefale, and Y. Gelaye, "Application of precision agriculture technologies for sustainable crop production and environmental sustainability: A systematic review," *Sci. World J.*, vol. 2024, art. no. 2126734, 2024, doi: 10.1155/2024/2126734.

[9] A. Abdullah, E. I. Raharjo, M. Iwan, R. Faizal, and Maryogi, "Development of an IoT-based soil nutrient monitoring and GIS mapping system for precision agriculture," *Adv. Sustain. Sci., Eng. Technol.*, vol. 7, no. 4, 2025.

[10] A. Erler *et al.*, "Mobile laser-induced breakdown spectroscopy for future application in precision agriculture—A case study," *Sensors*, vol. 23, no. 16, art. no. 7178, 2023, doi: 10.3390/s23167178.

[11] R. Kasemi, "EdgeSoil: Low-cost soil sensor and edge AI fusion in agricultural robotics," in *Proc. Austrian Robot. Workshop (ARW)*, Linz, Austria, 2023, pp. 1–6.

[12] M. J. Page *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, art. no. n71, 2021, doi: 10.1136/bmj.n71.

[13] B. Kitchenham and S. Charters, "Guidelines for performing systematic literature reviews in software engineering," Keele Univ. and Univ. of Durham, EBSE Tech. Rep. EBSE-2007-01, 2007.

[14] M. L. Rethlefsen *et al.*, "PRISMA-S: An extension to the PRISMA statement for reporting literature searches in systematic reviews," *Syst. Rev.*, vol. 10, art. no. 39, 2021, doi: 10.1186/s13643-020-01542-z.

[15] Uganda Bureau of Statistics, "Annual Agricultural Survey 2019: statistical release," Kampala, Uganda, 2022. [Online]. Available: https://www.ubos.org/wp-content/uploads/2022/04/Uganda_UBOS_StatRelease_AAS2019-Final.pdf

[16] Ministry of Finance, Planning and Economic Development, Uganda, "Agro-industrialisation: Agriculture remains the backbone of Uganda's economy," Knowledge Centre summary of the 2024 National Population and Housing Census, 2025. [Online]. Available: development.finance.go.ug

[17] Uganda Bureau of Statistics and Ministry of Agriculture, Animal Industry and Fisheries, "Census of Agriculture 2008–2009," FAO Food and Agriculture Microdata Catalogue. [Online]. Available: https://microdata.fao.org/index.php/catalog/1594

[18] J. L. Safanelli *et al.*, "Open Soil Spectral Library (OSSL): Building reproducible soil calibration models through open development and community engagement," *PLoS ONE*, 2025, doi: 10.1371/journal.pone.0296545.

[19] ISRIC—World Soil Information, "Africa Soil Profiles Database, version 1.2," ISRIC Data Hub. [Online]. Available: https://data.isric.org/geonetwork/srv/api/records/b88870b4-6af8-4e78-a3ac-38871d757525

[20] ISRIC—World Soil Information, "Standardised soil profile data for the world (WoSIS snapshot, September 2019)," doi: 10.17027/isric-wdcsoils.20190901.